

8(a)	<i>four from:</i> <i>thick filaments (max 3)</i> 1 myosin ; 2 fibrous protein ; 3 globular heads / ATPase ; 4 AVP ; e.g. 15 nm diameter / M lines <i>thin filaments (max 3)</i> 5 actin ; 6 globular protein ; 7 tropomyosin / troponin ; 8 binding site for myosin (head) ; 9 AVP ; e.g. 7 nm diameter / Z lines	4
8(b)	<i>five from:</i> 1 (when) sarcoplasmic reticulum depolarised ; 2 calcium (ion) <u>channels</u> open ; 3 calcium ions, move in (sarcoplasm) ; 4 bind to troponin ; 5 troponin / tropomyosin, changes shape ; 6 moves tropomyosin out of the way / exposes (actin-myosin) binding sites ; 7 (allows) myosin to bind (to actin) / cross bridge formation ;	5

Question	Answer	Marks
8(c)	<i>four from:</i> <i>description</i> 1 (pyruvate) accepts hydrogen / becomes reduced ; 2 from reduced NAD ; 3 produce lactate ; 4 ref. to lactate dehydrogenase ; <i>importance</i> 5 glycolysis can continue ; 6 ref. to small amount of ATP produced ; 7 AVP ; e.g. cannot continue for long / reversible / pyruvate does not enter mitochondrion	4